LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Control system components
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Control system components.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe various Aircraft engineering systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of control system diagrams; Safety: Do not operate live hydraulic systems; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft control system?
(b) Name three major categories of aircraft engineering systems.
(c) How does the control system differ from the power plant system?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Hydraulic power supply units • Control valves and actuators • Pilot control surfaces linkage • Feedback mechanisms in controls • System pressure monitoring devices	L & Ds	PPT, IB	15 min
(b)	• Electrical control actuation systems • Mechanical cable and pushrod systems • Hydro-mechanical servo systems • Redundancy in control circuits • Failure modes of control components	L & Ds	PPT, PDF	20 min
(c)	• Integration of flight control computers • Maintenance access points for controls • Standard torque values for control bolts • Inspection intervals for control linkages • Troubleshooting common control faults	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of hydraulic actuators in moving control surfaces.
(b) Describe one method used to ensure redundancy in the control system.
(c) Identify two common failure modes in mechanical cable systems.

12. Home Assignment and Follow Up	Prepare a brief note on the need for simple oxygen systems and the effects of lack of O2, including types of O2 regulators, maintenance, servicing, and safety, as covered in LP 10.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Need for simple oxygen system and pressurized cabin effect of lack of O2, Types of O2 regulators Maintenance of O2 system, servicing and safety precautions.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min